

Complex Analysis Qualifying Exam Review

Fall 2026

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Chapter 1

Fall 2025

Problem 1.1 (Theorem 5.2, Stein, Chapter 2). Let $\{f_n\}$ be a sequence of holomorphic functions on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that $f_n \rightarrow f$ uniformly on compact subsets of $\mathbb{D}$, where $f : \mathbb{D} \rightarrow \mathbb{C}$ is continuous. Prove that f is holomorphic on $\mathbb{D}$.

Proof. This is an immediate consequence of Morera's theorem. Take an arbitrary triangle T contained in $\mathbb{D}$, we know for any n ,

$$\int_T f_n(z) dz = 0$$

by Goursat's theorem. By assumption, $f_n \rightarrow f$ uniformly, thus

$$\int_T f(z) dz = \lim_{n \rightarrow \infty} \int_T f_n(z) dz = 0$$

By Morera's theorem, f is holomorphic in $\mathbb{D}$.

Morera's theorem. □

Problem 1.2. 6 Use contour integration to evaluate

$$\int_0^\infty \frac{x^2}{x^4 + 1} dx.$$

Proof. Use residue theorem for $z_1 = e^{\frac{\pi i}{4}}, z_2 = e^{\frac{3\pi i}{4}}$. □

Problem 1.3. 7 Let f be an analytic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \geq \sqrt{|z|}$$

for all $z \in \mathbb{D}$. Show that

$$|f(z)| \geq 1 \quad \text{for all } z \in \mathbb{D}.$$

Proof. Maximum modulus principle. □

Problem 1.4.8 Determine how many solutions

$$z^7 - 7z^4 + z^3 - 3z = 1$$

are in the annulus

$$1 < |z| < 2.$$

Proof. Rouché's theorem.

□

Chapter 2

Cauchy's integral formula

Proposition 2.1 (Cauchy's theorem). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U , if γ is a closed smooth curve whose interior is contained in U , then

$$\int_{\gamma} f(z) dz = 0$$

Proposition 2.2 (Cauchy's integral formula). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then for any $z \in D_r(z_0)$,

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw$$

where $\gamma = z_0 + re^{i\theta}$, $\theta \in [0, 2\pi]$. Moreover,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z)^{n+1}} dw$$

Proposition 2.3 (Cauchy's inequality). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then

$$|f^{(n)}(z_0)| \leq \frac{n! \sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.4 (Liouville's theorem). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a bounded entire function, then f must be constant.

Proposition 2.5 (Holomorphic and analytic). Let $f : U \rightarrow \mathbb{C}$ be holomorphic in U which contains the closure of a disk $D_r(z_0)$, for some $z_0 \in U$. Then f has a power series expansion at z_0 :

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D_r(z_0)$, where

$$a_n = \frac{f^{(n)}(z_0)}{n!}$$

This combined with the previous theorem gives a bound on the coefficients:

$$|a_n| \leq \frac{\sup_{|z|=r} |f(z)|}{r^n}$$

Proposition 2.6 (Morera's theorem). Suppose f is a continuous function in an open set U , if for any triangle T contained in D ,

$$\int_T f(z) dz = 0$$

then f is holomorphic.

Problem 2.1. Evaluate the following integral:

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

Proof. The answer is

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}$$

We use the contour of a semicircle skipping $x = 0$. Let $f(z) = \frac{1 - e^{iz}}{z^2}$, then by Cauchy's integral formula, we have

$$\int_{\gamma_R} f(z) dz + \int_{\gamma_\varepsilon} f(z) dz + \int_{|z| > \varepsilon} f(z) dz = 0 \quad (1)$$

where

$$\lim_{\varepsilon \rightarrow 0} \int_{|z| > \varepsilon} f(z) dz = 2 \int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

Hence it suffices to compute the first two terms in (1). Let $z = Re^{i\theta}$, we see that

$$\begin{aligned} \left| \int_{\gamma_R} f(z) dz \right| &= \left| \int_0^\pi \frac{1 - e^{iRe^{i\theta}}}{R^2 e^{2i\theta}} iRe^{i\theta} d\theta \right| \\ &\leq \int_0^\pi \frac{2}{R} d\theta \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. Similarly, we can compute $\int_{\gamma_\varepsilon} f(z) dz$ by setting $z = \varepsilon e^{i\theta}$ and letting $\varepsilon \rightarrow 0$.

$$\begin{aligned} \int_{\gamma_\varepsilon} f(z) dz &= \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon^2 e^{2i\theta}} i\varepsilon e^{i\theta} d\theta \\ &= i \int_\pi^0 \frac{1 - e^{i\varepsilon e^{i\theta}}}{\varepsilon e^{i\theta}} d\theta \rightarrow -\pi \end{aligned}$$

as $\varepsilon \rightarrow 0$, since

$$\lim_{a \rightarrow 0} \frac{1 - e^{ia}}{a} = -i$$

□

Problem 2.2. Let U be a region. Suppose f is holomorphic in U , and a sequence $\{z_n\} \subset U$ converges to some $z \in U$. If $f(z_n) = 0$ for all n , show that $f \equiv 0$.

Proof. Since $z_n \rightarrow z$, we know $f(z) = \lim_{z_n \rightarrow z} f(z_n) = 0$. Assume f is not identically zero, then writing it as a power series, we have

$$f(w) = (w - z)^n g(w)$$

where $g(w) = a_n + a_{n+1}(w - z) + a_{n+2}(w - z)^2 + \dots$, for some $a_n \neq 0$. We know $g(z) = a_n \neq 0$, hence there exists a neighborhood V around z such that $g(w) \neq 0$ whenever $w \in V$. Hence, if $w \in V$, then $f(w) = 0$ iff $w = z$. This is a contradiction to $z_n \rightarrow z$ and $f(z_n) = 0$ for all n . □



Warning 2.1. This shows that a nonzero holomorphic function has isolated zeros.

Problem 2.3 (Spring 2018, Q4). Let f be an entire function, suppose that for each z_0 , the power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has at least one coefficient $a_n = 0$. Show that f must be a polynomial.

Proof. It suffices to show that for some $k \in \mathbb{N}$, $f^{(k)} \equiv 0$ because

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We first observe that

$$\bigcup_{n \in \mathbb{N}} \{z : f^{(n)} = 0\} = \mathbb{C}$$

Then there must exist some $k \in \mathbb{N}$ such that

$$\{z : f^{(k)} = 0\} \text{ is uncountable}$$

Thus $f^{(k)}$ has uncountably many zeros, which implies $f \equiv 0$.

Lemma 2.1. Let f be a nonzero holomorphic function, then f cannot have uncountably many zeros.

Proof. Let $Z = \{z \in \mathbb{C} : f(z) = 0\}$, we know that Z contains no limit point. We can tile the complex plane using unit squares and suppose Z is uncountable, the intersection of Z with one of the tiles needs to be uncountable. This is a contradiction because this intersection is compact and discrete, hence must be finite. □

□

Problem 2.4 (Fall 2019, Q2). Let f be an entire function such that

$$\sup_{|z|>R} |f(z)| \leq AR^k + B$$

for positive constants A, B and all R large. Then show f is a polynomial with degree at most k .

Proof. This problem is a direct application of Cauchy's inequality. Writing $f(z)$ as a power series around 0, we have

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

It suffices to show that for any $m > k$, we have

$$f^{(m)}(0) = 0$$

Using Cauchy's inequality, we get

$$\begin{aligned} |f^{(m)}(0)| &\leq \frac{m! \sup_{|z|=R} |f(z)|}{R^m} \\ &\leq \frac{m!(AR^k + B)}{R^m} \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$ because $m > k$. Hence we are done. $\square$

Problem 2.5 (Fall 2023, Q1). (a) Show that if an entire function f satisfies

$$|f(z)| \leq \sqrt{|z|}$$

for all $z \in \mathbb{C}$, then $f(z)$ is a constant function.

(b) Why isn't the function $z^{1/2}$ a counterexample for Part (a)?

Proof. (a) We write f as the Taylor series about 0,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

Using Cauchy's inequality, for any R large, if $n \geq 1$,

$$|f^{(n)}(0)| \leq \frac{n! \sup_{|z|=R} |f(z)|}{R^n} \rightarrow 0$$

as $R \rightarrow \infty$. Thus $f(z) = a_0$, for some $a_0 \in \mathbb{C}$.

(b) The function is not entire. $\square$

Problem 2.6 (Fall 2018, Q4). Suppose f, g are both entire functions such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f = cg$.

Proof. We will use Liouville's theorem. It suffices to show that $h = f/g$ continuously extends to an entire function. Suppose $g(z_0) = 0$ for some z_0 , then we know $f(z_0) = 0$, hence writing

$$f(z) = (z - z_0)^n s(z), \quad g(z) = (z - z_0)^m t(z)$$

where $s(z_0) \neq 0, t(z_0) \neq 0$. We must have $n \geq m$, suppose if not, i.e., $m > n$, then

$$|s(z)| \leq |z - z_0|^{m-n} |t(z)|$$

which is a contradiction by letting $z = z_0$. Thus z_0 is a removable singularity for $h = f/g$. This implies h can be extended to an entire function, as desired. $\square$

Problem 2.7 (Spring 2015, Q6). Suppose f is a nonconstant entire function, then show that $f(\mathbb{C})$ is dense in $\mathbb{C}$.

Proof. Suppose there exists $w \in \mathbb{C}$ such that for a small $r > 0$, $D_r(w) \cap f(\mathbb{C}) = \emptyset$. Then we consider the function

$$\frac{1}{f(z) - w}$$

This function is entire and bounded, implying f is constant, which is a contradiction. $\square$

Chapter 3

Singularities

Definition 3.1 (singularities).

Definition 3.2 (residue). If f has a pole of order n at z_0 , then

$$\operatorname{res}_{z_0} f = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z - z_0)^n f(z)$$

Proposition 3.1 (Removable singularity). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a removable singularity at z_0 iff

$$\lim_{z \rightarrow z_0} (z - z_0) f(z) = 0$$

iff there exists a neighborhood N of z_0 where f is bounded on N .

Proposition 3.2 (Pole). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$. We say f has a pole at z_0 iff

$$\lim_{z \rightarrow z_0} |f(z)| = \infty$$

iff there exists a non-vanishing holomorphic function h and a unique n such that

$$f(z) = (z - z_0)^{-n} h(z)$$

Proposition 3.3 (Residue theorem). Suppose f is holomorphic in an open set containing a smooth closed curve γ and its interior, except for poles at points $z_1, \dots, z_n$ inside of γ . Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{i=1}^n \operatorname{res}_{z_i} f$$

Proposition 3.4 (Casorati-Weierstrass). Let f be a holomorphic function defined on $D_r(z_0) \setminus \{z_0\}$ and suppose f has an essential singularity at z_0 , then the image of f is dense in $\mathbb{C}$.

Proposition 3.5 (Argument principle).

Proposition 3.6 (Rouche's theorem).

Problem 3.1. Compute

$$\int_0^{\infty} \frac{1}{1+x^2} dx$$

Problem 3.2. Compute the following integral: fix $0 < a < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x} dx$$

Proof. Use rectangles and the residue at πi . □

Problem 3.3. Assume that $f : \mathbb{C} \rightarrow \mathbb{C}$ is an entire function, not identically equal to 0, and let

$$Z = \{z \in \mathbb{C} : f(z) = 0\}.$$

Prove that if Z is unbounded, then f has an essential singularity at ∞ .

Problem 3.4. Suppose f is an injective entire function, show f is linear.

Problem 3.5 (Spring 2017, Q6). Let $f : U \rightarrow \mathbb{C}$ be a non-constant holomorphic function where $U \subset \mathbb{C}$ is an open set containing the closure $\overline{D}$ of the unit disk

$$D = \{z \in \mathbb{C} : |z| < 1\}.$$

If $|f(z)| = 1$ for all $z \in \partial D$, then prove that

$$D \subset f(\overline{D}).$$

Proof. Argument+maximum modulus. □

Problem 3.6 (Fall 2015, Q4). Let $U \subset \mathbb{C}$ be an open set containing $\overline{D(P, r)}$. Prove that if $f : U \rightarrow \mathbb{C}$ is a holomorphic function such that f is nowhere zero on $\partial D(P, r)$ and $g : U \rightarrow \mathbb{C}$ is a holomorphic function sufficiently uniformly close to f on $\partial D(P, r)$, then the number of zeros of f in $D(P, r)$ equals the number of zeros of g in $D(P, r)$ (counting multiplicity).

Proof. Argument. □

Problem 3.7 (Spring 2021, Q1). Suppose $U \subset \mathbb{C}$ is an open set. Suppose f and g are functions holomorphic on $U \subset \mathbb{C}$. Show that if

$$|f(z)|^4 + |g(z)|^4$$

attains a maximum for some $z_0 \in U$, then f and g are constant on U .

Proof. MMP. □

Chapter 4

Conformal mapping

Definition 4.1 (automorphisms of the disk). Let $a \in \mathbb{D}$, the following function is an automorphism of the disk:

$$\phi_a(z) = \frac{z - a}{1 - \bar{a}z}$$

We have $\phi_a \circ \phi_{-a} = \text{id}$. Moreover, if f is a holomorphic function on $\mathbb{D}$, then $g(0) = 0$ if we let

$$g = \phi_{f(a)} \circ f \circ \phi_{-a}$$

Proposition 4.1 (Open mapping theorem). If $\Omega \subset \mathbb{C}$ is open and connected and $f : \Omega \rightarrow \mathbb{C}$ is a nonconstant holomorphic function, then f maps open sets to open sets.

Proposition 4.2 (Maximum modulus principle).

Proposition 4.3 (Schwarz's lemma).

Proposition 4.4 (Corollary of Schwarz's lemma).

Proposition 4.5 (Riemann mapping theorem).

Problem 4.1 (Spring 2019, Q4). Show that the punctured unit disk

$$\{z : 0 < |z| < 1\}$$

and the annulus

$$\{z : 1 < |z| < 2\}$$

cannot be conformally equivalent.

Problem 4.2 (Fall 2019, Q4). Let f be a holomorphic function in the punctured disk

$$\{z : 0 < |z| < 2\}$$

satisfying

$$|f(z)| \leq \left(\log \frac{1}{|z|} \right)^{100} \quad \text{in } \{z : |z| \leq 1/2\},$$

and

$$|f(z)| = 1 \quad \text{on } |z| = 1.$$

- (a) Show that f has a removable singularity at the origin.
- (b) Show that if $f(z) \neq 0$ in $|z| < 1$, then f is constant.

Problem 4.3 (Fall 2019, Q4). Let $f(z)$ be a holomorphic function on

$$\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\},$$

with $|f(z)| < 1$, and suppose $f(\alpha) = 0$ for some $|\alpha| < 1$. Show that for $z \in D$,

$$|f(z)| \leq \left| \frac{z - \alpha}{1 - \bar{\alpha}z} \right|.$$

Proof. Schwarz lemma. □

Problem 4.4 (Spring 2016, Q5). Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be a holomorphic function where

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

is the unit disk. Prove that if f has at least 2 fixed points then f is the identity map.

Proof. Let $z_1, z_2 \in \mathbb{D}$ be such that $f(z_1) = z_1, f(z_2) = z_2$. Then we consider the composite

$$g = \phi_{z_1} \circ f \circ \phi_{-z_1}$$

It is clear that g fixes 0, and if we let $z'_2 \neq 0$ be such that $\phi_{-z_1}(z'_2) = z_2$, then it is clear that g fixes z'_2 as well. We see that

$$g(0) = 0, \quad |g(z)| \leq 1, \quad g(z'_2) = z'_2$$

hence applying the Schwarz lemma, there exists $|c| = 1$ such that $g(z) = cz$, which implies g is constant. This shows f is constant as well. □

Chapter 5

Spring 2026

Problem 5.1. Let f be a holomorphic function on $\{z : |z| < 1\}$ that extends to a continuous function on $\{z : |z| \leq 1\}$ and maps the closed unit disk to itself. Furthermore, suppose

$$f(0) = f'(0) = \dots = f^{(n-1)}(0) = 0.$$

Show that:

(1)

$$|f(z)| \leq |z|^n \quad \text{for all } |z| \leq 1.$$

(2) If there is some $P \in D(0, 1) \setminus \{0\}$ such that

$$|f(P)| = |P|^n$$

or if

$$|f^{(n)}(0)| = n!,$$

then

$$f(z) = e^{i\theta} z^n.$$

Problem 5.2 (Also Fall 2023, Q2). Let $-1 < a < 0$. Compute

$$\int_0^\infty \frac{x^a}{1+x} dx.$$

Problem 5.3. Two parts.

(1) Let

$$f : U = \{z : 0 < |z| < 1\} \longrightarrow \mathbb{C}$$

be holomorphic and nonconstant. If there exists a sequence $\{z_n\} \subset U$ that converges to 0 and satisfies

$$f(z_n) = 0,$$

then prove that 0 is an essential singularity of f .

(2) Show that if f is an entire function that satisfies

$$\lim_{|z| \rightarrow \infty} f(z) = \infty,$$

then f is a polynomial.

Problem 5.4. Describe all injective holomorphic maps that map

$$\mathbb{C} \cup \{\infty\} \quad \text{to} \quad \mathbb{C} \cup \{\infty\}$$

and also map

$$\mathbb{R} \cup \{\infty\} \quad \text{to} \quad \mathbb{R} \cup \{\infty\}.$$

Proof. Let f be such a function, then by the Open Mapping Theorem, f is open. Now viewing $\mathbb{C} \cup \{\infty\}$ as the (compact) Riemann sphere, $f(\mathbb{C})$ is both open and closed, thus must be an automorphism. Since f also fixes $\mathbb{R} \cup \{\infty\}$, let $\mathbb{H}$ denote the upper half plane, then either $\mathbb{H} \mapsto \mathbb{H}$ or $\mathbb{H} \mapsto -\mathbb{H}$. In the first case, we know f must take the form

$$f(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{R}, ad - bc = 1$. In the second case, f can be written as the composite of the function $z \mapsto -z$ and a mobius transformation. Combined, all the maps are of the form

$$\left\{ \frac{az + b}{cz + d} : a, b, c, d \in \mathbb{R}, ad - bc = \pm 1 \right\}$$

□